VENUGOPAL INDRAJITH HEMASHREE

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EDUCATION

Vellore Institute of Technology, Chennai Aug 2023 – Expected 2027
B.Tech, Computer Science and Engineering (Cyber-Physical Systems) | CGPA: 7.81 / 10

Maharishi International Residential School, Kanchipuram XII: 79% | X: 89%

PROJECTS

TriageIQ — Real-Time LLM Support-Ticket Triage | GitHub 2026
Python, Apache Kafka, Ollama (qwen2.5), Qdrant, FastAPI, SQLite, Docker Compose

- Built an **event-driven pipeline** that turns incoming support emails (IMAP) into **Kafka** events, classifies category, priority and sentiment with a **local LLM**, and drafts replies grounded in similar past tickets retrieved from **Qdrant (RAG)**.
- Kept a **human in the loop**: a **FastAPI + Server-Sent Events** agent console streams tickets live, and replies go out over SMTP only after approval; invalid model output is routed to a person instead of being dropped.
- Hardened for real email: **Message-ID de-duplication**, **per-sender rate limiting**, auto-reply loop protection and signed sessions with lockout; runs **fully locally**, so no customer data leaves the machine.

RootLens — Automated Root-Cause Analysis for Microservices | GitHub 2026
Python, FastAPI, Docker Compose, Prometheus, Grafana, SQLite, GitHub Actions

- Built a **three-service FastAPI** system with **fault-injection** switches and a **Prometheus**-based detector that opens an incident after 3 checks at **30%+ failure ratio** and resolves it automatically on recovery.
- Wrote an investigator that scores each failing dependency as down, erroring or slow using up-metrics, error ratios and p95 latency, and returns **ranked root causes** with confidence levels and the **downstream services affected**.
- Added versioned releases with **rollback** and a policy-limited **remediation controller** (dry-run/enforce, action allowlist, rate limits) that **rolled back a faulty release automatically** in a live test; **CI** runs unit tests and image builds on every push.

RL-Driven Autonomous Rover for Disaster-Response Reconnaissance 2026
Python, Raspberry Pi 5, BreezySLAM, YOLOv8n / YOLOv9t, A, Reinforcement Learning, Flask*

- Ran **LiDAR SLAM** (BreezySLAM) on a **Raspberry Pi 5** to map the area in real time, overlaid with a **risk heatmap** of hazard zones from an MQ-2 gas sensor and a camera-based **YOLOv9t** model.
- Added a **risk-score head to YOLOv8n** and reweighted its loss toward **human detection**; a modified **A* planner** finds the safest exit path, and **RL** re-plans the sweep around new hazards.

AI Dental Malocclusion Detection with Explainable AI | GitHub Jul 2025 – Sep 2025
Python, PyTorch, YOLOv9, Grad-CAM, OpenCV, Gradio

- Trained a **YOLOv9** detector for **31 dental condition classes** on panoramic X-rays, reaching **97.1% mAP@50** (96.8% precision, 95.9% recall) on a held-out test set.
- Added **Grad-CAM** explanations, FDI tooth charting, PDF reports and a **Gradio** app for real-time use.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML/CSS
Backend & Streaming: FastAPI, REST APIs, Server-Sent Events, Apache Kafka, SQLite, MySQL
DevOps & Observability: Docker, Docker Compose, GitHub Actions (CI), Prometheus, Grafana, Git, Linux
AI / ML: LLMs (Ollama), RAG, Qdrant, PyTorch, TensorFlow, YOLOv8/v9, OpenCV, Grad-CAM, MMPose
Embedded / IoT: Raspberry Pi, Arduino, sensor fusion, LiDAR SLAM
Spoken: Tamil (native), English (fluent), Hindi (intermediate), Japanese (basic)

PATENT

IoT Smart Walker with Sensor-Fusion Fall-Risk Prediction — Co-Inventor Filing in progress
Inventive disclosure filed with IIPRD (Khurana & Khurana) | Faculty guide: Dr. Subitha D | Arduino, MPU6050, HX711, HC-SR04

- Designed a **fall-risk engine** and a **Dynamic Stability Reserve** metric that fuse bilateral tilt, weight asymmetry and obstacle distance to flag deteriorating stability before a fall, with **near-fall detection** and a **Gait Symmetry Index** for rehabilitation.
- Runs fully standalone on an **Arduino Uno** with no cloud dependency, at an estimated hardware cost **under USD 30**.

RESEARCH EXPERIENCE

Research Intern — VIT Chennai (Prof. Pavithra) | 1 month May 2025 – Jul 2025
Human Pose Estimation using Heatmaps | MMPose, OpenCV, Kinovea, Matplotlib, Python

- Implemented human joint detection with **heatmap-based pose estimation** models in **MMPose**, producing keypoints for each body joint.
- Visualised and analysed **joint-confidence heatmaps** with OpenCV and Matplotlib to study how well each keypoint was localised and where predictions became unreliable.
- Validated predicted keypoints against **Kinovea** motion-analysis measurements as ground truth, checking keypoint accuracy under faculty supervision.

CERTIFICATIONS & LEADERSHIP

- **IBM** GenAI with watsonx (Jun 2025, certificate) · **NUS** Industry 4.0 (Jun 2025, certificate) · **SuperAI 2025** (Singapore)
- **Team Leader**, CodeYGen Club (coding and design projects) · **Design Lead**, Fitness Club (event creatives)